

HOUSE PRICE PREDICTION USING LINEAR REGRESSION

Artificial Intelligence & Machine Learning – Task 1

Abstract

This project focuses on predicting California housing prices using a Linear Regression model. The complete machine learning workflow was implemented, including data loading, exploratory data analysis, preprocessing, model training, prediction, and performance evaluation. The California Housing dataset provides several socio-economic and geographical features which are used to estimate house prices. The objective is to demonstrate the application of supervised machine learning techniques for a real-world regression problem.

1. Introduction

Machine Learning enables computers to learn patterns from historical data and make predictions. Linear Regression is one of the most fundamental supervised learning algorithms used for predicting continuous values. In this project, Linear Regression is applied to the California Housing dataset to estimate median house prices based on demographic and geographic information.

2. Dataset Description

The California Housing dataset contains information collected from California census districts. Each record contains numerical attributes such as median income, house age, average rooms, average bedrooms, population, occupancy, latitude, and longitude. The target variable is the median house value. The dataset contains 20,640 observations and is widely used for regression tasks.

3. Project Objectives

- Load and explore the California Housing dataset.
- Perform exploratory data analysis (EDA).
- Build a Linear Regression model.
- Evaluate model performance using MAE, RMSE, and R^2 Score.
- Interpret results and identify future improvements.

4. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset structure and relationships among variables. The dataset was inspected using `head()`, `info()`, and `describe()` functions. Correlation analysis helps identify which features have the strongest influence on house prices. Histograms and scatter plots were used to visualize distributions and relationships.

Key Observations

- The dataset contains no missing values.
- Median income shows a strong positive relationship with house prices.
- Geographic location significantly influences property values.
- Most variables are numerical, making them suitable for regression analysis.

5. Methodology

Step 1: Load dataset using Scikit-Learn.
Step 2: Convert data into a Pandas DataFrame.
Step 3: Split data into training and testing datasets.
Step 4: Train the Linear Regression model.
Step 5: Generate predictions on unseen data.
Step 6: Evaluate model performance.

6. Model Training

The dataset was divided into training and testing subsets using an 80:20 ratio. The `LinearRegression()` model from Scikit-Learn was trained using the training data. During training, the model learned the relationship between input features and housing prices.

7. Evaluation Metrics

Mean Absolute Error (MAE): Measures the average prediction error.

Root Mean Squared Error (RMSE): Penalizes larger errors more strongly.

R² Score: Measures how well the model explains variability in the target variable.

Typical Results

MAE $\approx$ 0.53

RMSE $\approx$ 0.74

R² Score $\approx$ 0.57 – 0.60

8. Result Analysis

The model achieves moderate predictive performance. An R² score close to 0.60 indicates that approximately 60% of the variation in housing prices can be explained by the selected features. While Linear Regression provides a strong baseline model, more sophisticated algorithms may further improve prediction accuracy.

9. Future Improvements

- Feature Engineering
- Polynomial Regression
- Ridge and Lasso Regression
- Random Forest Regressor
- Gradient Boosting Models
- Hyperparameter Optimization

10. Conclusion

This project successfully demonstrated an end-to-end machine learning workflow using the California Housing dataset. The Linear Regression model was trained and evaluated effectively, providing valuable insights into housing price prediction. The project highlights the importance of data exploration, preprocessing, model evaluation, and result interpretation in real-world machine learning applications.